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M1 MACRO

Group B

Session 4

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

I need to solve

$$\frac{1}{c_1} = \beta(1+m)E \frac{1}{c_2} \quad \leftarrow$$

Remark $E =$ mathematical expectation
 $=$ linear operator

$$[E(ax+b) = aE(x) + b]$$

but

$$E\left[\frac{1}{x}\right] \neq \frac{1}{E(x)}$$

Assume $\beta(1+r)=1$

Euler eq: $\frac{1}{C_1} = E \left[\frac{1}{C_2} \right]$

Let us consider 2 situations

① stochastic: $y_2 = \begin{cases} y_L & \text{prob } 1/2 \\ y_H & \text{prob } 1/2 \end{cases}$
 $y_L < y_H$

$C_2 = y_2 + (1+r)S = \begin{cases} c_L & \text{prob } 1/2 \\ c_H & \text{prob } 1/2 \end{cases}$

(2) Deterministic

The agent gets $E(y_2)$ for sure

$$E(y_2) = \frac{1}{2} y_H + \frac{1}{2} y_L$$

so that

$$C_2 = E(C_2) = \frac{1}{2} C_H + \frac{1}{2} C_L$$

For $\frac{1}{C_1} = \underbrace{\beta(C_2)}_1 \frac{1}{C_2} \Leftrightarrow \frac{1}{C_1} = \frac{1}{E(C_2)}$

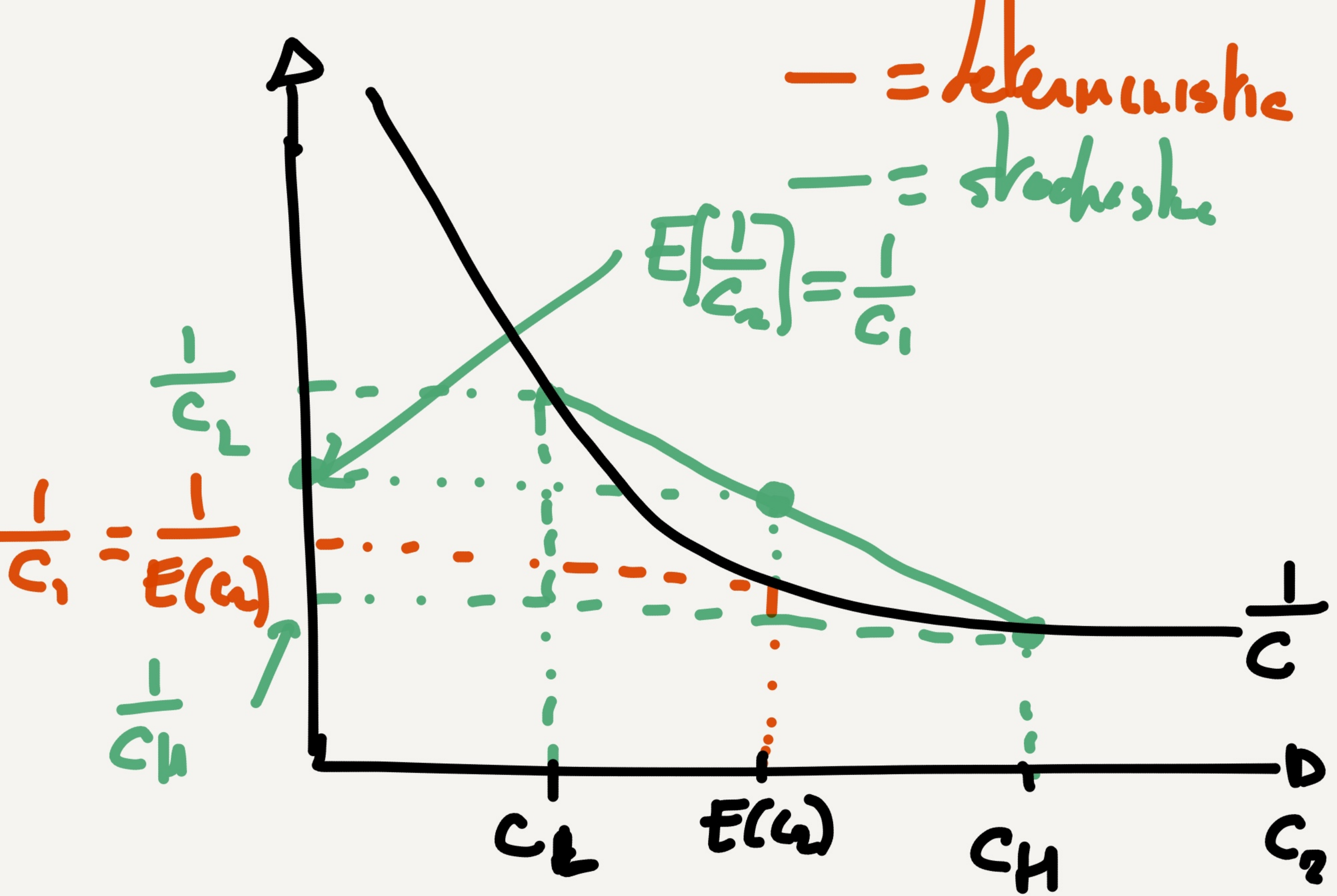
• Stochastic world

$$\frac{1}{C_1} = E \frac{1}{C_2}$$

• Deterministic world

$$\frac{1}{C_1} = \frac{1}{E C_2}$$

How the choice of C_1 today is affected by uncertainty?



$$\frac{1}{C_1} \text{ with uncert.} > \frac{1}{C_1} \text{ with certainty}$$

$$\Rightarrow C_1 \text{ uncert.} < C_1 \text{ certainty}$$

$$S = Y_1 - C_1$$

$$S_{\text{uncert.}} > S_{\text{certainty}}$$

PRECAUTIONARY SAVING

2. period model with labor supply

[not general equilibrium]

$$M = \log C_1 + \beta \log C_2 + \log(1 - N_1) + \beta \log(1 - N_2)$$

1 unit of time available per period

$N_1 =$ working time

$1 - N_1 =$ leisure time

$$0 \leq N_1 < 1$$

$$I \beta c : c_1 + \frac{c_2}{R} = w_1 N_1 + \frac{w_2 N_2}{R}$$

$$R = 1 + n$$

$$\begin{aligned} \mathcal{L} = & \log c_1 + \rho \log c_2 + \log(1 - N_1) + \rho \log(1 - N_2) \\ & + \lambda \left[w_1 N_1 + \frac{w_2 N_2}{R} - c_1 - \frac{c_2}{R} \right] \end{aligned}$$

choose optimally: c_1, c_2, N_1, N_2
 $\lambda > 0$

FOC

$$/c_1: \frac{1}{c_1} = \lambda$$

$$/c_2: \beta / c_2 = \lambda / R$$

$\Rightarrow$

$$c_1 = c_2 = c$$

$$/w_1: \frac{1}{1-N_1} = \lambda w_1$$

$\Rightarrow$

$$1-N_1 = \frac{c}{w_1}$$

$$N_1 = 1 - \frac{c}{w_1}$$

$$/N_2: \cancel{\beta} / \cancel{1-N_2} = \lambda \cancel{\frac{w_2}{R}}$$

$$1-N_2 = 1 - \frac{c}{w_2}$$

Assume $\beta R = 1$

plug it back in the TBC

$$C + \frac{C}{R} = w_1 \left(1 - \frac{C}{w_1}\right) + \frac{w_2}{R} \left(1 - \frac{C}{w_2}\right)$$

$$\Rightarrow \left(1 + \frac{1}{R}\right) C = w_1 + \frac{w_2}{R} - C - \frac{C}{R}$$

$\underbrace{\hspace{10em}}_{- C \left(1 + \frac{1}{R}\right)}$

$$\boxed{C = \frac{1}{2} \frac{R}{1+R} \left(w_1 + \frac{w_2}{R}\right)} = C_1 = C_2$$

$$N_1 = 1 - \frac{c}{w_1} = 1 - \frac{1}{2} \frac{R}{1+R} \left(1 + \frac{w_2}{w_1} \frac{1}{R} \right)$$

$$N_2 = 1 - \frac{c}{w_2} = 1 - \frac{1}{2} \frac{R}{1+R} \left(\frac{w_1}{w_2} + \frac{1}{R} \right)$$

① Temporary wage increase
 $\tilde{w}_1 = \lambda w_1 \quad \lambda > 1$

$$N_1 \nearrow, N_2 \searrow$$

② Permanent wage increase
 $\tilde{w}_1 = \lambda w_1, \tilde{w}_2 = \lambda w_2 \quad \lambda > 1$

No change in labor supply